

Multiple Choice

1. **Answer:** A

Options: A) Cu; B) Gd; C) Eu; D) Am

Note: Copper (Cu) has a completely filled 3 d orbital and a partially filled 4 s orbital, but it does not have partially filled 4 f or 5 f orbitals, distinguishing it from the other elements listed in the question.

2. **Answer:** B

Options: A) NH^3 ; B) NH_2^- ; C) NH_4^+ ; D) N_2H_4

Note: NH_2^- (amide ion) is the strongest base in liquid ammonia because it possesses a high electron density due to its negative charge, allowing it to readily donate electrons and accept protons more effectively than other species present in that solvent.

3. **Answer:** D

Options: A) racemic mixture; B) single pure enantiomer; C) mixture of two diastereomers in equal amounts; D) meso compound

Note: The reduction of D-xylose with NaBH_4 leads to the formation of a meso compound because the resulting product, D-xylitol, contains multiple stereocenters that are symmetrically arranged, resulting in an internal plane of symmetry that makes it achiral.

4. **Answer:** B

Options: A) 23.56 GHz; B) 42.58 MHz; C) 74.34 kHz; D) 13.93 MHz

Note: The Larmor frequency for a proton in a magnetic field of 1 T is calculated using the formula $f = \gamma B$, where γ (the gyromagnetic ratio for protons) is approximately 42.58 MHz/T, thus resulting in a frequency of 42.58 MHz when B is 1 T.

5. **Answer:** D

Options: A) electron distribution; B) dipole moment; C) force constant; D) reduced mass

Note: The infrared absorption frequency of deuterium chloride (DCl) is shifted from that of hydrogen chloride (HCl) because the reduced mass, which depends on the masses of the atoms involved, affects the vibrational frequency of the molecular bond, leading to differences in their absorption spectra.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: Strong oxidizing conditions are not required for paramagnetism and ferromagnetism, as these magnetic properties arise from the presence of unpaired electrons in

certain materials, which can occur under various conditions regardless of oxidation state.

7. **Answer:** True

Options: A) True; B) False

Note: The molecular geometry of thionyl chloride (SOCl_2) is trigonal pyramidal because the presence of a lone pair on the sulfur atom results in a tetrahedral electron geometry that is distorted by the lone pair, leading to a pyramidal shape.

8. **Answer:** False

Options: A) True; B) False

Note: Tetramethylsilane is actually used as a reference in NMR spectroscopy because its protons are highly shielded, providing a stable and consistent reference point for chemical shifts.

9. **Answer:** True

Options: A) True; B) False

Note: Cr_3^+ has an electronic configuration of $[\text{Ar}] 3d^3$, which results in three unpaired electrons, making it paramagnetic and capable of acting as a quencher.

10. **Answer:** False

Options: A) True; B) False

Note: Infrared spectroscopy is not a light-scattering technique; instead, it measures the absorption of infrared light by chemical bonds in a substance, making it a technique based on light absorption rather than scattering.

Long Form Response

11. **Answer:** To balance the redox reaction between permanganate ions (MnO_4^-) and iodide ions (I^-), we find that the stoichiometric ratio of MnO_4^- to I^- is 6:5. This ratio arises from the fact that each MnO_4^- ion is reduced to Mn^{2+} in an acidic medium, consuming a total of 8 electrons, while each I^- ion is oxidized to I_2 , releasing 2 electrons. Therefore, for every 6 moles of MnO_4^- that are reduced, 5 moles of I^- are oxidized, leading to the balanced equation. This 6:5 ratio highlights the specific electron transfer and conservation of mass principles in redox reactions, illustrating how stoichiometry reflects the underlying chemical processes involved.

Note: correct because it accurately describes the electron transfer process, indicating that 6 moles of MnO_4^- are reduced (gaining 8 electrons each) for every 5 moles of I^- oxidized (losing 2 electrons each), resulting in the stoichiometric ratio of 6:5 in the balanced redox reaction.

12. **Answer:** The Heisenberg uncertainty principle has significant implications for a quantum mechanical particle in the lowest energy level of a one-dimensional box, particularly regarding the limitations in precisely determining its position and momentum. According to this principle, the more accurately we know the position of the particle, the less accurately we can know its momentum, and vice versa. In the lowest energy

state, the particle exhibits a wave-like behavior, described by a probability distribution, which inherently limits our ability to pinpoint its exact location. Consequently, we can only describe the particle's position and momentum in terms of probabilities rather than definitive values, highlighting the fundamental nature of uncertainty in quantum mechanics. This intrinsic limitation challenges classical notions of determinism and emphasizes the probabilistic nature of quantum systems.

Note: correct because it accurately describes the Heisenberg uncertainty principle's effect on a quantum particle's position and momentum, emphasizing that the wave-like nature of the particle in its lowest energy state leads to inherent limitations in precisely measuring these properties simultaneously.

End of Answer Key